

GeneyMAP: Designing GenAI-empowered User Journey Mapping Tool

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Abstract

Despite the growing integration of Generative AI (GenAI) in UX design, its role in facilitating Journey Map (JM) creation remains insufficiently explored. In this paper, we focus on the development process of GeneyMAP, a GenAI-empowered design tool that aims to streamline the JM creation process. Starting with a formative study, we identified eight essential functions necessary for common JM creation. These functions were further refined through a co-design session involving six UX design experts, who were all familiar with GenAI applications, focusing on effectively integrating GenAI to optimize these functions within the JM creation workflow. This collaboration led to the design of the functional prototype of GeneyMAP. This study delineates a replicable process for embedding GenAI in design tools, offering insights into its effective integration into the JM creation process.

CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**.

Keywords

Generative AI, LLMs, Design Tool, UX Design, User Research, Prompt Engineer, Journey Map

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1 Introduction

Journey map (JM), a pivotal user experience (UX) design tool, visually outlines user interactions with a service or product, guiding designers in crafting user-centered experiences [34]. Despite its utility, common JM tools face challenges in managing extensive data and depend largely on manual qualitative analysis, which significantly hinders UX designers from efficiently identifying user pain points and design opportunities, thereby delaying the final design delivery. Under these circumstances, the need for innovation in the JM creation process becomes imperative.

Generative artificial intelligence (GenAI), represented by tools like ChatGPT and DALL-E, has rapidly advanced in generating textual and visual content. With its ability to process large datasets [5], derive insights [27, 30], and create visualizations [2, 35], GenAI offers transformative potential for UX design [40]—a central focus of Human-Computer Interaction (HCI) research aimed at improving user interactions with products or systems [21]. Based on that, the GenAI's strengths in synthesizing information [29, 31, 36], identifying opportunities [7], and stimulating creativity [1, 9] presents significant potential to improve the JM creation process. HCI researchers have actively explored the integration of GenAI to improve UX design-fostering creativity [39], supporting collaborative design [16], and in particular by optimizing traditional tools such as personas [31] and storyboards [17]. However, systematically using GenAI to streamline JM creation while maintaining workflow integrity remains an open question [36].

In addressing this gap, our study investigated the application of GenAI in streamlining the JM creation process, proposing a repeatable process for designing GenAI-empowered design tools. Based on a formative study with 24 UX designers and a co-design process involving six experts, we introduced a system called **GeneyMAP**, which refined the JM creation process by incorporating eight tailored GenAI-empowered functions. These functions were optimized

through a co-design process to effectively embed GenAI within the JM creation workflow. While this work primarily focuses on the co-design process and technical aspects of GeneyMAP's development, the evaluative user study compared GeneyMAP with a common JM creation tool was also conducted in [19]. Both quantitative and qualitative results indicate that GeneyMAP outperformed the common JM tool in several dimensions, including the JM creation process and the quality of JM outputs. However, weaknesses were also found in terms of transparency, trust, and control [19]. The development of GeneyMAP not only demonstrates the practical application of GenAI in design but also serves as a blueprint for future advancements in the field, highlighting the significant enhancements it brings to the efficiency and creativity of design processes and design implications for integrating GenAI into the design of future design tools.

2 BACKGROUND

JMs depict user interactions temporally, providing insights into behavior patterns critical to UX analysis [15], extensively employed across educational and corporate settings. The JM process initiates with diverse data collection approaches, including interviews, surveys, and observational research [11, 34], followed by mapping user **scenarios**, interaction **phases**, and other **elements** such as touchpoints, actions, thoughts, goals and emotion [8, 33]. The process identifies actionable insights to enhance UX by pinpointing **pain points** and **opportunities** [20, 37]. Challenges in managing dynamic datasets and interpreting complex user data underscore the need for improved JM tools.

The HCI community recognizes GenAI's potential to revolutionize JM by enhancing productivity [10, 13] and creativity [18, 28]. GenAI aids in automating early UX stages [4], analyzing data, and optimizing design layouts through applications like ChatGPT and DALL-E, which assist in persona creation [14, 29] and prototyping [5, 12, 24, 25]. However, GenAI's challenges, such as producing hallucinations and logical inconsistencies [3, 6, 39, 41], and its dependence on prompt quality [18], pose limitations. Addressing these, This study proposes a GenAI-empowered JM tool that integrates automated insights with manual customizations, maintaining authenticity while leveraging AI's capabilities [26].

3 THE DESIGN OF GENEYMAP

3.1 Formative Study

To determine the essential functions of a JM creation tool, we conducted semi-structured interviews with 24 design-trained participants from diverse industries and academic backgrounds. Drawing on their experiences with existing tools (e.g., Figma, Miro, Xmind) with JM creation, participants offered valuable insights into the necessary features. An inductive thematic analysis of the interview data revealed several critical functions, which are presented in the following section as **F1–F8**, outlining the key functions that an effective JM creation tool should include (see Figure 1.a).

F1- Identifying data patterns. This function analyzes raw user data from interviews, surveys, or observations to uncover recurring themes and behaviors. It systematically extracts key patterns to establish a structured foundation for the JM process.

F2-Mapping user information. Building on identified data patterns, this function organizes insights into a journey framework, detailing user actions, thoughts, emotions, and goals in a sequential format. It aims to provide a clear and coherent representation of user interactions.

F3-Formulating supplementary information. This function enriches the JM with inferred elements like emotions and unspoken needs, enhancing understanding of the user experience by capturing subtle and indirect user data aspects.

F4-Reasoning pain points. Following the mapping and supplementary information, this function identifies and prioritizes moments of user frustration or unmet needs, classifying pain points by their impact on the user experience to guide improvements.

F5-Ideating design opportunities. This function transforms identified pain points into actionable design opportunities, focusing on creating solutions that address specific user frustrations highlighted earlier in the process.

F6-Supporting content iteration. This function facilitates the continuous refinement of JMs, updating pain points and opportunities as new data or ideas emerge, ensuring the map's relevance and accuracy over time.

F7-Enhancing language expression. This function improves the clarity, precision, and coherence of textual descriptions within the JM, ensuring effective communication and fostering collaboration among stakeholders.

F8-Evidencing. This function ensures that the mapped information faithfully represents the user's exact expressions and intended meanings.

3.2 The Co-design Process

To develop a comprehensive and practical solution aligned with UX designers' expectations and real-world needs, GeneyMAP was created using a co-design approach. This approach involved pilot tests and iterative refinements to ensure the effectiveness of the system.

Before the pilot tests, we designed a **draft functional prototype**, which deliberately diverged from the standard JM creation workflow to stimulate innovative thinking. In typical JM creation workflows, designers begin by collecting data, such as conducting user interviews. However, our prototype reverses this order, first establishing a template and then involving users to fill it in based on their own experiences. This approach is designed to fully leverage GenAI's data analysis capabilities, allowing structured user input to enhance the efficiency and effectiveness of the analysis process. Moreover, rather than simplifying the design by eliminating features, we chose to incorporate all the interesting functions we envisioned in combination with GenAI at this design stage. For instance, a clipboard feature was designed to store generated texts and images at the end of the process, but it was later deemed unnecessary. It is important to note that only the functions closely related to **F1–F6** (e.g., reveal pain points from filled JMs) have been implemented in a preliminary, functional form, while other features (e.g., the clipboard function) were added to the interface solely to explore their potential without full functionality. To assess the feasibility of these initial design choices and identify necessary refinements, we conducted pilot tests with design experts.

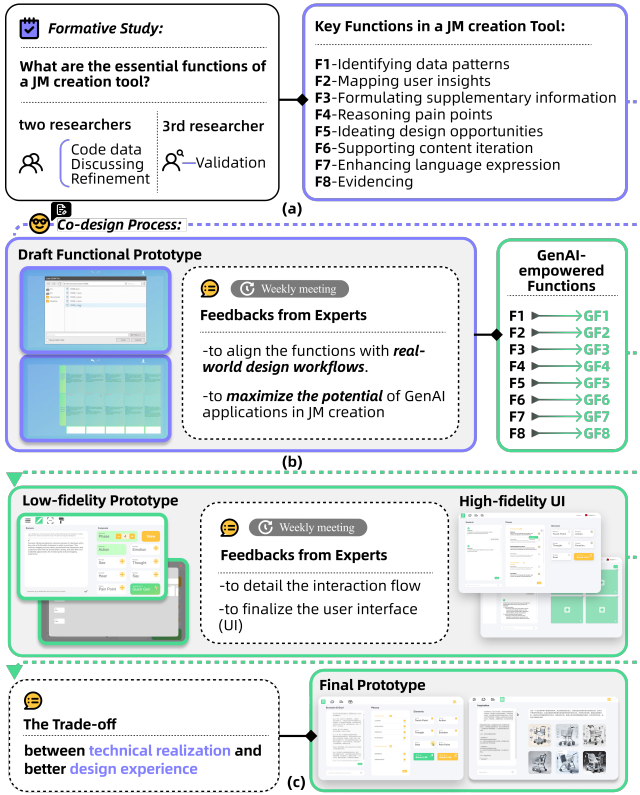


Figure 1: The design process of GeneyMAP prototype.

3.2.1 Pilot Tests. The **draft functional prototype** served as a foundation to refine the content and scope of the functions themselves. Six design experts from diverse professional backgrounds, including two practitioners, two researchers, and two educators, were recruited to contribute their expertise. These experts, with 6 to 12 years of journey mapping experience and more than a year of using GenAI to support design, brought varied levels of proficiency in GenAI—from consistent integration in design projects to frequent usage. The diversity of their expertise facilitated a well-rounded and comprehensive refinement of the co-design process, ensuring the functions were purposefully tailored to practical design needs. Experts were introduced to the draft functional prototype through a workflow demonstration and an accompanying video, which provided the necessary context for their conceptual understanding and hands-on evaluation. Weekly evaluation meetings were conducted to analyze the feedback, focusing on improvements in function clarity, workflow efficiency, and applicability.

3.2.2 Iteration Refinement of Functions. Expert feedback from the pilot tests indicated that altering the common JM creation workflow introduced too much cognitive shift for designers, making the tool less intuitive and harder to adopt. They also considered changing the workflow solely to accommodate GenAI capabilities as a misguided approach. Ideally, GenAI should be integrated into the existing workflow to enhance efficiency while maintaining the quality of outputs. This approach would allow designers

to fully utilize their expertise and reduce the cost of adopting a GenAI-empowered JM creation tool. Ultimately, a tool’s value is determined by its practical usability.

Based on expert recommendations derived from their engagement with the draft functional prototype, the original functions (**F1-F6**) for common JM creation were refined with a clear design rationale aimed at *aligning them with real-world design workflows and optimizing their potential for GenAI applications in journey map creation*. Through iterative expert feedback and functional refinement, we expanded the original F1–F6 functions into an upgraded set of eight GenAI-empowered functions (**GF1-GF8**) that better addressed the practical needs of JM creation (see Figure 1.b), the detailed description of these functions is shown in Table 1.

With the refined functions established, development transitioned to a **low-fidelity prototype**, which built upon the draft functional prototype by incorporating basic interaction flows and user interface elements. This prototype aimed to optimize usability and seamlessly integrate the refined functions into a more tangible and interactive design. Throughout this phase, regular weekly meetings with experts were maintained to facilitate the gradual implementation of the **high-fidelity UI** into the functional prototype. Close collaboration with developers was crucial during this stage, involving the creation of one-to-one demonstration visuals and conducting offline meetings for iterative design refinement. These refinements extended beyond the user interface to include adjustments to the system prompt, ensuring alignment with evolving project objectives. Given the system’s non-commercial nature and time constraints, we prioritized essential interactions over secondary features, ultimately leading to the development of a **high-fidelity functional prototype** (see Figure 1.c). Key enhancements at this stage included a restructured visual hierarchy, improved user interaction flows, and better alignment with the refined functions.

The high-fidelity prototype underwent a heuristic evaluation by experts to validate the seamless integration of the new eight functions and the overall usability of the system. This iterative process, guided by pilot testing and expert collaboration, ensured that GeneyMAP emerged as a robust and efficient tool tailored to the evolving needs of UX designers, offering a comprehensive solution for JM creation.

4 GeneyMAP IMPLEMENTATION

GeneyMAP is a GenAI-empowered tool designed to accelerate JM creation while preserving real-world design workflows. Constructed on the Unity platform, GeneyMAP employs OpenAI’s GPT-4o [23] and DALL-E 3 [22] APIs to perform specialized textual analysis and generate visual representations for JM creation through advanced prompt engineering. A standardized JSON format is utilized to streamline data transfer and precisely render UI components within Unity. Furthermore, the system’s comprehensive UI management offers features such as undo capabilities, and collapsible panels. Moreover, the Runtime File Browser [38] plugin streamlines file management within Unity. Additionally, guidance from RageAgainstThePixel-OpenAI [32] underpins a robust and

GenAI-empowered Function	GenAI-empowered Function Description
GF1: Identifying the scenario, phases, and elements	GeneyMAP automates the analysis of interview transcripts to extract scenarios, phases, and elements like touchpoints, actions, and emotions. It uses semantic analysis to identify patterns, enabling designers to refine and validate these insights.
GF2: Automating user information mapping	This function automates the translation of interview data into structured JMs. Designers can refine generated insights, merge overlapping data, and ensure alignment with the predefined JM template.
GF3: Generating supplementary information	GeneyMAP simulates synthetic user journeys based on existing scenarios to provide alternative perspectives, enriching the insights while maintaining core data integrity.
GF4: Reasoning pain points	By leveraging LLM-driven reasoning, this function identifies and highlights user pain points from JMs, offering triggers for generating actionable design opportunities.
GF5: Ideating and visualizing design opportunities	Designers can generate actionable design opportunities directly from identified pain points. Each trigger produces opportunity statements with visual suggestions, ensuring alignment with user needs.
GF6: Supporting collaborative content iteration	GeneyMAP integrates an interactive GenAI agent to refine design concepts collaboratively. The agent maintains context awareness and simplifies complex ideas, enhancing team communication and iterative development.
GF7: Enhancing language expression	This function improves the clarity and quality of textual descriptions, ensuring consistency and precision across the journey mapping process for effective communication.
GF8: Evidencing	During the co-design phase, experts unanimously expected this function to be entirely managed by humans, and as a result, no GenAI applications were designed to support the evidence process.

Table 1: Refined GenAI-empowered functions in GeneyMAP.

secure GenAI integration. Therefore, GeneyMAP offers a comprehensive solution to meet the evolving needs of JM creation. The following are the details of constructing the GeneyMAP.

4.1 The Design Mechanism of GeneyMAP System

To address the need for a reasoning mechanism to support the specified GenAI-empowered functions (*GF1-GF8*), GeneyMAP employs a progressive **four-step creation mechanism** based on the Chain of Thought (CoT) methodology.

- **Step 1. Distilling the Structured JM Template:** Establish a foundational JM framework by processing interview scripts to extract the underlying scenario, key phases, and core elements, which designers then refine into a complete JM template.
- **Step 2. Generating Individual JMs:** Convert raw interview data into individual JMs that adhere to an established scenario and template. Additionally, an optional synthetic user feature offers alternative insights for enriched perspectives.
- **Step 3. Merging JMs and Identifying Opportunities:** This step provides functions to consolidate individual journey maps into a comprehensive overview by identifying and

eliminating overlaps. It helps uncover valuable pain points and generates diverse design opportunities with visual suggestions.

- **Step 4. Exploring Design Inspirations:** Exploring Design Inspirations Enhances the JM process through an interactive agent that provides iterative design feedback, supports concept refinement, and crafts high-quality visual and textual outputs.

Specifically, by integrating the CoT approach with the foundational principles of JM, these principles are translated into LLM prompts, forming the basis of the four-step creation mechanism operational workflow of GeneyMAP. For instance, these principles include defining the roles and tasks assigned to GenAI, formalizing key JM terminologies (e.g., scenario, phases, touchpoints, actions, emotion, pain points, and opportunities), and outlining the structured process for JM creation. Moreover, to mitigate potential randomness in AI-generated content that could deviate from user expectations, the output format for each step in the four-step creation mechanism has been standardized, establishing a tightly coupled relationship between the steps. Furthermore, illustrative examples are embedded within the rule-based prompts as constraints

to further refine and control the generated content (see the sample prompts in Appendix A). The overview of the GeneyMAP JM creation workflow is presented in Figure 2.

Notably, the four-step creation mechanism incorporates a "human-in-the-loop" approach, fostering a tight human-AI collaboration. While the mechanism automates most JM creation tasks, each step allows designers to intervene in real-time, enabling them to leverage their expertise to make more informed design decisions. This collaborative mechanism ensures that GeneyMAP produces professional and practical JM content that aligns with designers' final design objectives. The detailed eight refined functions (GF1-GF8) of each step in the mechanism are described below.

4.2 Implementation of the GenAI-empowered Functions at Each Step

Step 1. Distilling the Structured JM Template. This step establishes the groundwork for JM creation by implementing **GF1**. It utilizes prompt engineering to extend GPT-4o's semantic analysis capabilities and applies statistical methods for structured data processing.

The process begins when designers upload interview data through GeneyMAP's interface (see Figure 2.a). Using advanced prompt engineering, the system instructs GPT-4o to analyze the transcripts by applying semantic grouping and key concept recognition. For example, prompts guide the model in identifying and grouping similar user statements, classifying actions, emotions, and events, and establishing recurring themes. This process employs techniques like topic modeling to extract recurring patterns from the text, forming the basis for auto-generating initial scenarios. Scenarios generated from this process are refined via interactive GenAI prompts or manual adjustments. Designers finalize these scenarios, which are embedded as system prompts to guide subsequent steps. Statistical methods such as leveraging the relative distribution and significance of terms across the dataset, are employed to stratify data segments into "common" and "uncommon" phases, enabling designers to review and validate these categorizations. The horizontal axis of the JM template is constructed from these phases, while designers define vertical elements (e.g., touchpoints, actions, emotions) to complete the template. JSON synchronizes the structured template data with Unity's interface, ensuring accurate visualization and dynamic updates.

Step 2. Generating Individual JMs. This step converts raw user data into structured JMs, enhancing insights through additional perspectives (see Figure 2.b). It implements **GF2** and **GF3**, employing GPT-4o for semantic mapping and synthetic user generation.

Uploaded transcripts are processed through custom-designed prompts, instructing GPT-4o to perform contextual analysis by mapping user-specific insights into the JM template. Prompts are designed to extract key "element data", such as actions and emotions, by using sentence-level sentiment analysis and attention-based phrase extraction. For instance, phrases indicating user goals ("I wanted to...") or pain points ("It was difficult to...") are tagged and aligned with the corresponding phases in the template. Each transcript produces one JM, ensuring fidelity to the original user experience. The synthetic user function utilizes scenario-driven

prompts to simulate alternative behaviors and profiles. By employing probabilistic language modeling and persona-based embeddings, GPT-4o generates supplementary JMs that simulate realistic user journeys. For instance, in a healthcare setting, synthetic personas such as patients with different conditions are generated. JSON facilitates the integration of these outputs into Unity for visualization, where designers can refine them interactively using collapsible sections and GenAI-driven text optimization.

Step 3. Merging JMs and Identifying Opportunities. This step consolidates individual JMs into an overview and generates design opportunities from selected pain points (see Figure 2.c). It implements **GF4** and **GF5**, leveraging GPT-4o's reasoning algorithms and DALL-E 3 for visualization.

The "Merge" function utilizes prompt engineering to guide GPT-4o in identifying overlaps across individual JMs. Techniques like semantic similarity measures and embedding-based comparisons are applied to align and cluster similar insights into unified categories. For example, statements like "It's hard to find options" and "Navigation is confusing" are grouped into a common theme representing a navigation-related pain point. Reasoning prompts further instruct GPT-4o to analyze the root causes of these pain points by examining associated phases and actions. Once pain points are identified, GPT-4o generates two opportunity statements per point through creative prompts designed to ensure contextual relevance. Visual suggestions for these opportunities are generated using DALL-E 3, guided by descriptions derived from opportunity statements. For example, a navigation-related pain point might generate an opportunity like "Introduce an intuitive navigation menu" with a corresponding visual of a simplified UI design. Unity's interface allows designers to refine these opportunities interactively, supported by undo and dynamic update functions.

Step 4. Exploring Design Inspirations. This step supports iterative refinement and ideation, enabling designers to produce actionable outputs while enhancing team collaboration. It implements **GF6**, utilizing GenAI-driven interactive agents (see Figure 2.d).

Contextual prompts activate GPT-4o to assist designers in refining design concepts. The prompts guide the model in reorganizing complex design ideas into concise, actionable suggestions using summarization techniques and sentiment refinement. For example, designers can ask the agent to simplify a detailed user scenario into key insights or to brainstorm alternative approaches to solve specific pain points. GPT-4o uses advanced semantic understanding and contextual analysis to iteratively refine the phrasing and coherence in descriptions of the opportunities and provide inspiring stimuli. DALL-E 3 is also integrated for visual ideation, with image prompts crafted from refined opportunity statements. These outputs are dynamically displayed in Unity, allowing for real-time designer interaction and validation.

Furthermore, **GF7** was integrated in all four steps as a basic function to refine JM descriptions for clarity and consistency. The prompt pseudocode for implementing the eight GenAI-empowered functions in the four Steps is shown in Appendix B.

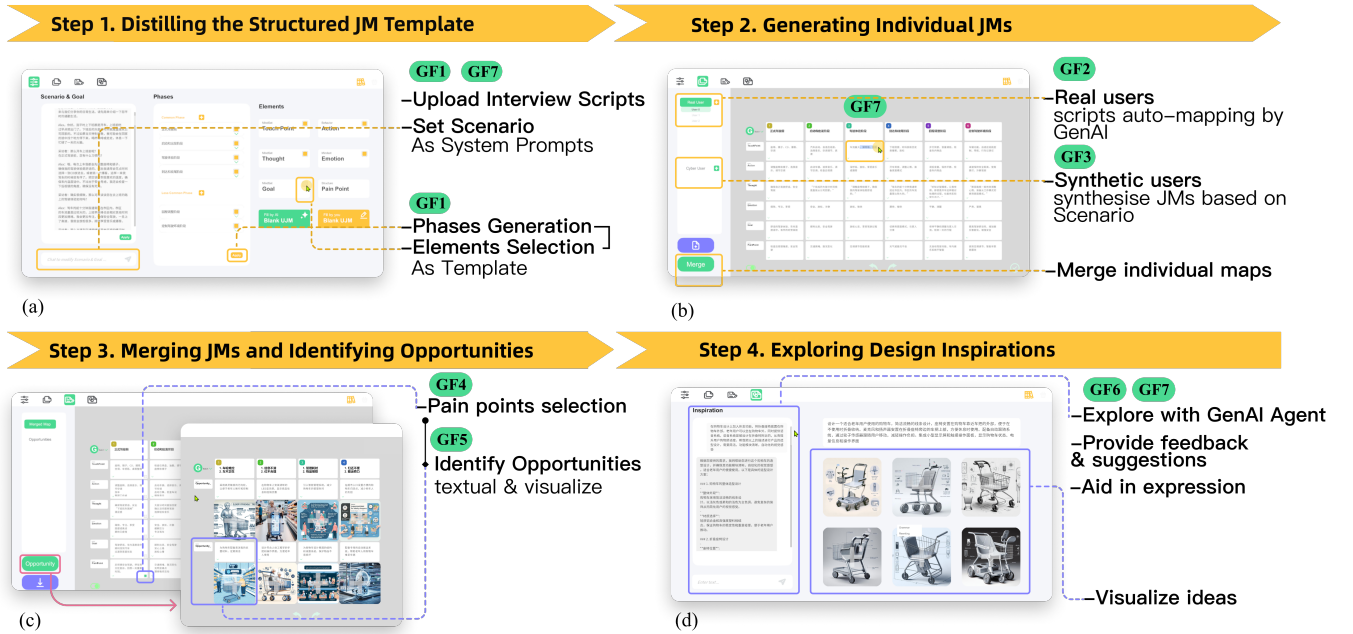


Figure 2: GeneyMAP workflow empowered GF1-GF8 in four steps

5 DISCUSSION

5.1 Insights from the co-design process of GeneyMAP

This section explores two central themes that emerged during our co-design process. First, we discuss the challenges of balancing varied GenAI automation levels across the eight JM creation functions. GenAI is expected to accelerate the JM creation process, yet determining the optimal automation level is a difficult trade-off because excessive automation can compromise output quality. Second, we reflect on how designer feedback—grounded in their experiences with the GeneyMAP functional prototype and our unconventional method—provided critical insights, enabling us to refine the balance between automated assistance and human expertise.

Challenges in balancing varied GenAI automation levels. Our co-design process revealed that the eight functions integral to JM creation require distinct levels of GenAI automation and assistance. For example, functions like *F1 - Identifying data patterns* and *F2 - Mapping user information* can benefit from high degrees of automation through semantic analysis and structured data extraction. However, excessive automation in these areas may lead to misclassification of nuanced user inputs. Conversely, *F8 - Evidencing* demands minimal automation, as experts unanimously agreed that this function should remain entirely human-managed to ensure fidelity to users' exact expressions. Similarly, *F3 - Formulating supplementary information* and *F5 - Ideating design opportunities* illustrate the need for a balanced approach—GenAI can generate synthetic insights and opportunity statements, yet human intervention is crucial for refining and contextualizing these outputs. This heterogeneity in automation requirements makes it challenging to

determine the precise balance between AI assistance and human oversight within the existing JM creation workflow.

Prioritizing designer feedback in GenAI integration. Although our research focuses on UX and user research, the primary users of these tools are the designers. Feedback from designers during the co-design sessions was pivotal in navigating these trade-offs. By engaging with a functional prototype that deviated from common existing JM creation methods, designers were encouraged to think more critically and innovatively, avoiding clichéd solutions. This insight guided our decision to refine the functions to align better with established design practices while still exploring the potential of GenAI. For example, experts in our study noted that modifying the common JM creation workflow solely to leverage GenAI imposed an excessive cognitive load on designers. They argued that GenAI should be integrated into the existing workflow to enhance efficiency while preserving the designer's role and expertise. Listening to designer feedback is crucial, as it ensures that GenAI serves as a collaborative partner that enhances creativity and decision-making, rather than as a disruptive force.

5.2 Implications for GenAI-empowered UX Design Tools

The integration of GenAI-empowered functions into JM creation presents significant opportunities to enhance efficiency and insight generation. However, achieving meaningful integration would require addressing challenges related to workflow adaptation and the balance between automation and human input. A strategic approach is necessary to ensure that GenAI enhances, rather than disrupts, the design process.

A co-design approach is critical for achieving effective integration. By involving UX designers early and consistently, the development of GenAI-empowered functions can align more closely with real-world design needs. Co-design fosters iterative improvements, enabling designers to refine AI-generated outputs based on contextual knowledge and domain expertise. This collaborative process helps ensure that AI suggestions remain relevant and actionable within diverse design scenarios.

While GeneyMAP primarily addresses JM creation, its underlying co-design methodology offers valuable insights that can be applied to a broader range of GenAI-empowered design tools. Ensuring human oversight is crucial to mitigate the risk of over-reliance on AI-generated outputs. Rather than replacing designer intuition, GenAI should be leveraged as a collaborative partner that enhances human creativity and decision-making. Establishing clear guidelines on the capabilities and limitations of GenAI functions is essential to promote responsible use and uphold the integrity of the design process.

5.3 Limitations and Future Considerations

Several limitations of GeneyMAP should be acknowledged. First, due to time and technical constraints, the study did not explore more advanced GenAI-empowered implementations, which may yield better expert evaluations and enhance JM creation, warranting further investigation. Second, the current UX and UI design may not be fully optimized, potentially affecting participant experiences in future user studies and influencing the reliability of findings. Refining the interface is essential to ensure effective tool adoption. Third, the co-design process involved a limited number of experts familiar with GenAI, which may not fully capture diverse user needs. Future studies should include a broader participant base to improve generalizability. Lastly, the tool's predefined workflow may limit flexibility. Exploring more adaptable and customizable AI functions could better accommodate varying design requirements.

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APPENDIX - A

GenAI.GetPhase

```

Role.System
//①
Define the role, task, and key terms to avoid ambiguity in the system's output.
role = "Professional UX Designer"
task = "Summarize user journey phases based on interview scripts."
key_terms = {
  "phases": "Steps in the journey capturing significant milestones or shifts in interaction.",
  "common_phases": "Frequently occurring and similar phases across users.",
  "less_common_phases": "Rarely occurring or unique phases."
}

Emphasize structured and step-by-step thinking to handle complex tasks.
Step 1: Identify chronological phases.
Step 2: Separate phases into CommonPhase and LessCommonPhase based on frequency.
Step 3: Output results in JSON format.

//②
Provide specific examples to guide the understanding of the terms and desired output structure.
examples = {
  "ecommerce_scenario": ["discover", "try", "buy", "use", "seek support"],
  "luxury_purchase": ["engagement", "education", "research", "evaluation", "justification"],
  "b2b_scenario": ["purchase", "adoption", "retention", "expansion", "advocacy"]
}

//③
System processes interview scripts and outputs phases in JSON format.
def summarize_phases(interview_scripts):
  Step 1: Parse scripts to extract phases in chronological order.
  phases = extract_phases(interview_scripts)

  Step 2: Analyze frequency of occurrence.
  common_phases = [phase for phase in phases if is_frequent(phase)]
  less_common_phases = [phase for phase in phases if not is_frequent(phase)]

  Step 3: Return structured output in JSON format.
  return {
    "CommonPhases": common_phases,
    "LessCommonPhases": less_common_phases
  }

Example input: User interview scripts.
interview_scripts = [
  "...users start by researching online, then compare options, visit stores, and finalize purchases.",
  "...customers first learn about the product through ads, research features, then buy and leave reviews."
]

Example output in JSON format:
output_phases = summarize_phases(interview_scripts)

Expected JSON Output:
{
  "CommonPhases": ["research", "compare options", "buy"],
  "LessCommonPhases": ["visit stores", "leave reviews"]
}

```

Figure 3: The sample prompts.

APPENDIX - B

```
function extract_scenario_and_phases(transcripts: List[str]) -> Dict:
    """
    Extracts recurring patterns, generates scenarios, and categorizes phases.
    """
    prompt = {
        "instruction": "Analyze transcripts to identify patterns and generate scenarios.",
        "input": transcripts,
        "constraints": ["Identify key actions, emotions, and phases."],
        "output_format": {
            "scenario": "Scenario text",
            "phases": {"common": "Frequent phases", "uncommon": "Rare phases"}
        }
    }
    return GPT_API.call(prompt)
```

Figure 4: Prompt pseudocode for implementing functions in Step.1.

```
function map_user_data_to_JM(transcript: str, template: Dict) -> Dict:
    """
    Maps user insights into a predefined JM template.
    """
    prompt = {
        "instruction": "Align user insights with the JM template.",
        "input": transcript,
        "template": template,
        "constraints": ["Ensure context relevance."],
        "output_format": {
            "JM": {"scenario": "Scenario text", "phases": "Mapped phases"}
        }
    }
    return GPT_API.call(prompt)

function generate_synthetic_JMs(scenario: str, num_simulations: int) -> List[Dict]:
    """
    Simulates user behaviors to create synthetic JMs.
    """
    prompt = {
        "instruction": "Generate alternative journey maps based on scenario.",
        "input": scenario,
        "constraints": ["Ensure diverse user behaviors."],
        "output_format": [{"synthetic_JM": "Generated JM"}]
    }
    return GPT_API.call(prompt)
```

Figure 5: Prompt pseudocode for implementing functions in Step.2.

```
function merge_JMs_and_extract_opportunities(JMs: List[Dict]) -> Dict:
    """
    Merges JMs, identifies pain points, and generates opportunities.
    """
    prompt = {
        "instruction": "Merge JMs, extract pain points, and suggest opportunities.",
        "input": JMs,
        "constraints": ["Ensure insights are validated."],
        "output_format": {
            "merged_JM": {"pain_points": "List of pain points", "opportunities": "Opportunity list"}
        }
    }
    return GPT_API.call(prompt)

function generate_visual_suggestions(opportunity: str) -> str:
    """
    Generates visuals for design opportunities.
    """
    prompt = {
        "instruction": "Create visual suggestions for opportunity.",
        "input": opportunity,
        "output_format": "Generated visual description"
    }
    return DALL_E_API.call(prompt)
```

Figure 6: Prompt pseudocode for implementing functions in Step.3.

```
function refine_design_idea(idea: str, context: str) -> str:
    """
    Refines design ideas based on feedback and context.
    """
    prompt = {
        "instruction": "Refine design idea for clarity and alignment with goals.",
        "input": {"idea": idea, "context": context},
        "output_format": "Refined idea text"
    }
    return GPT_API.call(prompt)

function enhance_text_content(content: str) -> str:
    """
    Improves clarity and coherence of textual content.
    """
    prompt = {
        "instruction": "Enhance the clarity of the provided content.",
        "input": content,
        "output_format": "Enhanced content text"
    }
    return GPT_API.call(prompt)
```

Figure 7: Prompt pseudocode for implementing functions in Step.4.